

SYS 304: Trade Studies, Risk, and Decision Analysis

Simulation Approaches for Decision Making

A Reliability Problem

A milling machine has three different bearings that fail in service. The distribution of the life of each bearing is identical (Table 1). When a bearing fails, the mill stops, a repairperson is called, and a new bearing is installed. The delay time of the repairperson's arriving at the milling machine is also a random variable having the distribution given in Table 2. Downtime for the mill is estimated at \$10 per minute. The direct on-site cost of the repairperson is \$30 per hour. It takes 20 minutes to change one bearing, 30 minutes to change two bearings, and 40 minutes to change three bearings. A proposal has been made to replace all three bearings whenever a bearing fails. Management needs an evaluation of the proposal. The total cost per 10,000 bearing-hours will be used as the measure of performance.

Bearing Life (hours)	Probability	Cumulative Probability
1000	0.10	0.10
1100	0.13	0.23
1200	0.25	0.48
1300	0.13	0.61
1400	0.09	0.70
1500	0.12	0.82
1600	0.02	0.84
1700	0.06	0.90
1800	0.05	0.95
1900	0.05	1.00

Delay Time (Minutes)	Probability	Cumulative Probability
5	0.6	0.6
10	0.3	0.9
15	0.1	1.0

Table 3 represents a simulation of 15 bearing changes under the current method of operation. Note that there are instances where more than one bearing fails at the same time. This is unlikely to occur in practice and is due to using a rather coarse grid of 100 hours for bearing life. It will be assumed in this example that the times are never exactly the same and thus no more than one bearing is changed at any breakdown.

	Bearing 1		Bearing 2		Bearing 3	
	Life (hours)	Delay (min)	Life (hours)	Delay (min)	Life (hours)	Delay (min)
1	1400	10	1500	10	1100	5
2	1300	5	1100	5	1100	5
3	1900	5	1500	5	1400	5
4	1500	5	1700	5	1000	10
5	1300	5	1800	15	1300	10
6	1400	10	1500	5	1100	5
7	1500	5	1000	10	1100	5
8	1800	10	1100	10	1000	10
9	1200	15	1100	5	1300	5
10	1000	5	1100	5	1600	15
11	1900	10	1000	5	1100	5
12	1400	5	1000	15	1200	5
13	1300	10	1200	5	1700	10
14	1700	5	1000	5	1400	5
15	1700	5	1100	5	1200	5
Total	22,300	110	18,700	110	18,600	105

The cost of the current system is estimated as follows:

Cost of bearing = 45 bearings * \$32/bearing = \$1,440

Cost of delay time = (110 + 110 + 105) minutes * \$10/minute = \$3,250

Cost of downtime during repair = 45 bearings * 20 minutes/bearing * \$10/minute = \$9,000

Cost of repairpersons = 45 bearings * 20 minutes/bearing * \$30/60minutes = \$450

Total cost = \$1,440+\$3,250+\$9,000+\$450 = \$14,140

The total life of the bearings is (22,300 + 18,700 + 18,600) = \$59,600 hours. Therefore, the total cost per 10,000 bearing-hours is (\$14,140/5.96) = \$2,372.

Table 4 is a simulation of the proposed method. Note that the random digits are not shown. For the first set of bearings, the earliest failure is at 1,000 hours. All three bearings are replaced at that time, even though the remaining bearings had more life in them. For example, Bearing 1 would have lasted 700 additional hours.

The cost of the proposed system is estimated as follows:

Cost of bearings = 45 bearings * \$32/bearing = \$1,440

Cost of delay time = 110 minutes * \$10/minute = \$1,100

Cost of downtime during repair = 15 sets * 40 minutes/set * \$10/minute = \$6,000

Cost of repairpersons = 15 sets * 40 minutes/bearing * \$30/60minutes = \$300

Total cost = \$1,440+\$1,100+\$6,000+\$300 = \$8,840

	Bearing 1 Life (hours)	Bearing 2 Life (hours)	Bearing 3 Life (hours)	First Failure (hours)	Delay (minutes)
1	1700	1100	1000	1000	10
2	1000	1800	1200	1000	5
3	1500	1700	1300	1300	5
4	1300	1100	1800	1100	5
5	1200	1100	1300	1100	5
6	1000	1200	1200	1000	10
7	1500	1700	1200	1200	5
8	1300	1700	1000	1000	10
9	1800	1200	1100	1100	15
10	1300	1300	1100	1100	5
11	1400	1300	1900	1300	10
12	1500	1300	1400	1300	5
13	1500	1800	1200	1200	10
14	1000	1900	1400	1000	5
15	1300	1700	1700	1300	5
Total				17,000	110

The total life of the bearings is $(\$17,000 \times 3) = 51,000$ hours. Therefore, the total cost per 10,000 bearing-hours is $(\$8,840/5.1) = \$1,733$.

The new policy generates a savings of \$634 per 10,000 hours of bearing-life. If the machine runs continuously, the savings are about \$556 per year.